

[PART I. TRAJECTORY]

< Introduction >

①

① Notion of Equivalence

Sets of Coordinates which can be transformed uniquely

② Dynamical EoM

$$\textcircled{1} \quad \frac{dx_i}{dt} = f_i(x_1, \dots, x_n, u_1, \dots, u_m), \quad i=1, \dots, n$$

$$\Leftrightarrow \frac{dx}{dt} = f(x, u) \quad f: \mathbb{R}^n \times \mathbb{R}^m \Rightarrow \mathbb{R}^n$$

② Composition of Initial Value Problem (IVP)

$$\frac{dx}{dt} = f(x, u) \quad \begin{cases} x(t_0) = x_0 \\ u(t) = \bar{u}(t) \text{ is given on } t_0 \leq t \leq t_f \end{cases}$$

③ Equilibrium Solutions ~~($\bar{x}$)~~ $(\bar{x}, \bar{u})$

$$f(\bar{x}, \bar{u}) \equiv 0, \quad t \geq t_0 \quad \text{for a given } u(t) = \bar{u}$$

④ Perturbation Dynamics

Supp. $(\bar{x}, \bar{u})$ is a pair of equilibrium sol.

$$\text{Define } \begin{cases} x = \bar{x} + \delta x \\ u = \bar{u} + \delta u \end{cases}$$

$$\textcircled{1} \quad \frac{dx}{dt} = \frac{d\bar{x}}{dt} + \frac{d\delta x}{dt}$$

$$\textcircled{2} \quad f(x, u) = f(\bar{x} + \delta x, \bar{u} + \delta u) \\ = f(\bar{x}, \bar{u}) + \frac{\partial f}{\partial x} \bigg|_{x=\bar{x}, u=\bar{u}} \delta x + \frac{\partial f}{\partial u} \bigg|_{x=\bar{x}, u=\bar{u}} \delta u + \mathcal{O}(\|\delta x\|, \|\delta u\|)$$

$$\textcircled{1} = \textcircled{2}$$

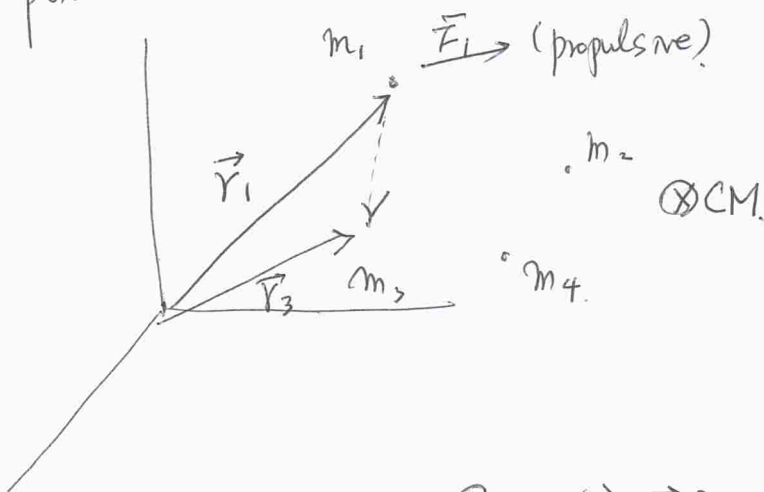
(2)

$$\frac{d\bar{x}}{dt} + \frac{d\alpha}{dt} = f(\bar{x}, \bar{u}) + \frac{\partial f}{\partial \alpha} \alpha + \frac{\partial f}{\partial u} \delta u + O(\|\alpha\|, \|\delta u\|)$$

$$\Rightarrow \boxed{\frac{d\alpha}{dt} = A\alpha + B\delta u + \text{H.O.T.}}$$

< N-Body Model >

⊙ Assumption: Point Mass (No attitude dynamics)



$$m_i \ddot{\vec{r}}_i = \sum_{j \neq i} \frac{G m_i m_j (\vec{r}_j - \vec{r}_i)}{|\vec{r}_i - \vec{r}_j|^3} + \vec{F}_i, \quad i=1, \dots, n$$

$$\Rightarrow m_i \ddot{\vec{r}}_i = \sum_{j \neq i} \frac{G m_i m_j}{|\vec{r}_{ij}|^3} \vec{r}_{ij} + \vec{F}_i, \quad i=1, \dots, n$$

⊙ Conservative System..

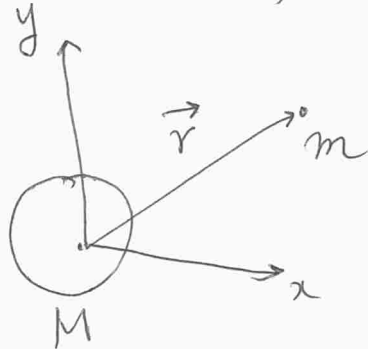
No external (= propulsive) forces $\Leftrightarrow \vec{F}_i \equiv 0$.

$$\sum_{i=1}^n m_i \ddot{\vec{r}}_i = 0$$

$$\Rightarrow \sum_{i=1}^n m_i \dot{\vec{r}}_i = C \quad (\text{Conservation of Linear Momentum})$$

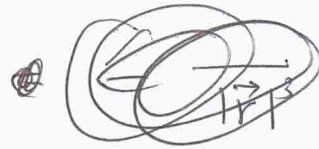
< 2-Body Model >

① 'Restricted' 2-body Model (in the Inertial Frame)



② ~~Inertial Coordinate~~

$$1) \ddot{\vec{r}} = -\frac{GMm}{|\vec{r}|^3} \vec{r} + \vec{F}$$



11) $\vec{F} \equiv 0 \Rightarrow$ free orbit described by traj. eqn.

$$r = \frac{p}{1 + e \cos \theta}$$

③ Cartesian Coordinates.

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\ddot{\vec{r}} = \ddot{x}\hat{i} + \ddot{y}\hat{j} + \ddot{z}\hat{k}$$

Let $\mu = GM$

$$\ddot{\vec{r}} = -\frac{\mu}{|\vec{r}|^3} \vec{r} + \frac{\vec{F}}{m}, \quad \vec{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}$$

$$\ddot{x} = \dots$$

(4)

 $\Rightarrow$ By Component

$$\begin{cases} \ddot{x} = -\frac{\mu x}{r^3} + \frac{F_x}{m} \\ \ddot{y} = -\frac{\mu y}{r^3} + \frac{F_y}{m} \\ \ddot{z} = -\frac{\mu z}{r^3} + \frac{F_z}{m} \end{cases} \quad r = \sqrt{x^2 + y^2 + z^2}$$

 $\Rightarrow$ State Space Form

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -\frac{\mu x_1}{r^3} + \frac{F_x}{m} \\ \dot{x}_3 = x_4 \\ \dot{x}_4 = -\frac{\mu x_3}{r^3} + \frac{F_y}{m} \\ \dot{x}_5 = x_6 \\ \dot{x}_6 = -\frac{\mu x_5}{r^3} + \frac{F_z}{m} \end{cases} \quad r = \sqrt{x_1^2 + x_3^2 + x_5^2}$$

$$x \triangleq \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} x \\ \dot{x} \\ y \\ \dot{y} \\ z \\ \dot{z} \end{bmatrix}$$

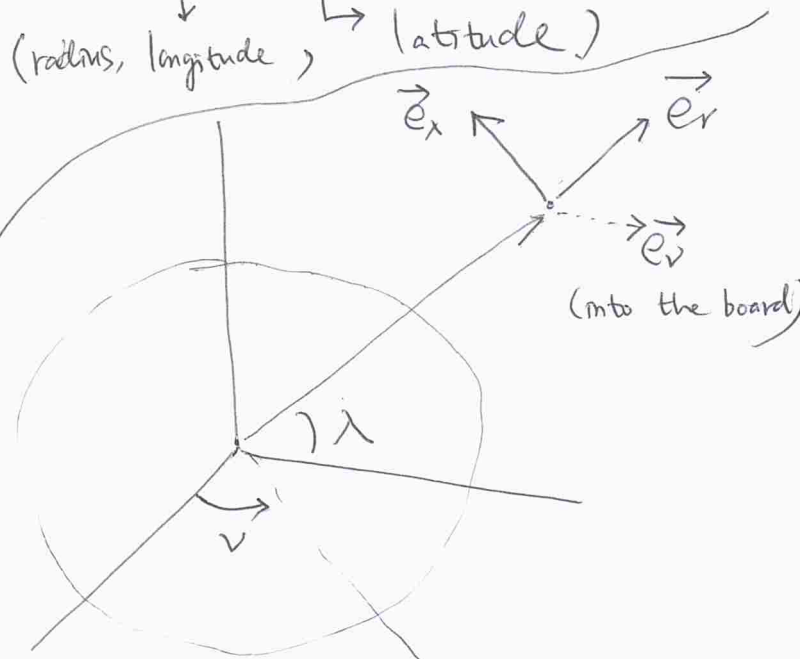
($\frac{x}{r}, \frac{y}{r}, \frac{z}{r}$)

© Transformation into Spherical Coord.

$$(x, y, z) \Leftrightarrow (r, \nu, \lambda)$$

$\Leftrightarrow$ (radius, longitude, latitude)

$$\begin{cases} x = r \cos \lambda \cos \nu \\ y = r \cos \lambda \sin \nu \\ z = r \sin \lambda \end{cases}$$



(5)

$$\vec{r} = r \vec{e}_r$$

$$\dot{\vec{r}} = \dot{r} \vec{e}_r + r \dot{\vec{e}}_r$$

$$= \dot{r} \vec{e}_r + r(\vec{\omega} \times \vec{e}_r) \quad (1)$$

$$\vec{\omega} = -\dot{\lambda} \vec{e}_v + \dot{v} \cos \lambda \vec{e}_\lambda + \dot{v} \sin \lambda \vec{e}_r \quad (2)$$

$$\therefore \dot{\vec{r}} = \dot{r} \vec{e}_r + r \dot{v} \cos \lambda \vec{e}_\lambda + r \dot{\lambda} \vec{e}_\lambda \quad (2 \Rightarrow 1)$$

$$\ddot{\vec{r}} = (\ddot{r} \vec{e}_r + \dot{r} \dot{\vec{e}}_r) + (\dots) + \dots$$

$$\ddot{\vec{r}} = [\ddot{r} \vec{e}_r + \dot{r}(\vec{\omega} \times \vec{e}_r)] + (\dots) + \dots$$

$$\Rightarrow \ddot{\vec{r}} = (\ddot{r} - r \dot{v}^2 \cos^2 \lambda - r \dot{\lambda}^2) \vec{e}_r + (r \ddot{v} \cos \lambda + 2 \dot{r} \dot{v} \cos \lambda - 2 r \dot{v} \dot{\lambda} \sin \lambda) \vec{e}_v + (r \ddot{\lambda} + 2 \dot{r} \dot{\lambda} + r \dot{v}^2 \sin \lambda \cos \lambda) \vec{e}_\lambda \quad (3)$$

$$\text{Gravity} \quad - \frac{\mu}{|\vec{r}|^3} \vec{r} = - \frac{\mu}{r^2} \vec{e}_r \quad (4)$$

$$\text{Propulsion} \quad \vec{F} = F_r \vec{e}_r + F_v \vec{e}_v + F_\lambda \vec{e}_\lambda \quad (5)$$

From (3) ~ (5)

$$\begin{cases} \ddot{r} - r \dot{v}^2 \cos^2 \lambda - r \dot{\lambda}^2 = - \frac{\mu}{r^2} + \frac{F_r}{m} \\ r \ddot{v} \cos \lambda + 2 \dot{r} \dot{v} \cos \lambda - 2 r \dot{v} \dot{\lambda} \sin \lambda = \frac{F_v}{m} \\ r \ddot{\lambda} + 2 \dot{r} \dot{\lambda} + r \dot{v}^2 \sin \lambda \cos \lambda = \frac{F_\lambda}{m} \end{cases} \quad (6)$$

~~Special Case~~

Special Case

i) If $\vec{F} \equiv 0$, then one solution is.

$$\times \begin{cases} \lambda \equiv 0 \\ \ddot{r} - r\dot{v}^2 = -\frac{\mu}{r^2} \\ r\ddot{v} + 2\dot{r}\dot{v} = 0 \end{cases} \quad \begin{matrix} \textcircled{6} - 3 \\ \textcircled{6} - 1 \text{ or } 2 \end{matrix}$$

ii) If $\vec{F} = 0$, then another solution is

$$\bigcirc \begin{cases} r \equiv R \\ v = \sqrt{\frac{\mu}{R^3}} t + C \\ \lambda \equiv 0 \end{cases}$$

which can be simply verified by substitution.

Linearization

Assume (r, v, λ) is not necessarily zero.

Let $(r=R, v=\sqrt{\frac{\mu}{R^3}}t, \lambda=0)$ be operating point.

$$\bigcirc \begin{cases} r = R + \Delta r \\ v = \sqrt{\frac{\mu}{R^3}}t + \Delta v \\ \lambda = 0 + \Delta \lambda \end{cases} \quad \textcircled{7}$$

(7)

~~Then introducing~~Then, (7) $\Rightarrow$ (6).

$$i) (\ddot{r} + \dot{r}) - (R + \Delta r) \left(\sqrt{\frac{\mu}{R^3}} + \Delta v \right)^2 \cos^2 \Delta \lambda - (R + \Delta r) (\Delta \lambda)^2$$

$$= - \frac{\mu}{(R + \Delta r)^2} + \frac{T_r}{m}$$

binomial expansion or Taylor Expansion

$$\Rightarrow \ddot{r} - R \left(2 \sqrt{\frac{\mu}{R^3}} \Delta v \right) - \Delta r \frac{\mu}{R^3} - \frac{\mu}{R^2} = - \frac{\mu}{R^2} + \frac{2\mu}{R^3} \Delta r + \frac{T_r}{m}$$

$$\Rightarrow \boxed{\ddot{r} - 2 \sqrt{\frac{\mu}{R^3}} R \Delta v - 3 \frac{\mu}{R^3} \Delta r = \frac{T_r}{m}} \quad (8) - 1.$$

$$ii) (R + \Delta r) \ddot{\Delta \lambda} \cos \Delta \lambda + 2 (\dot{r} + \dot{\Delta r}) \left(\sqrt{\frac{\mu}{R^3}} + \Delta v \right) \cos \Delta \lambda$$

$$- 2 (R + \Delta r) \left(\sqrt{\frac{\mu}{R^3}} + \Delta v \right) \dot{\Delta \lambda} \sin \Delta \lambda = \frac{T_v}{m}$$

$\sin \Delta \lambda \approx \Delta \lambda$

$$\Rightarrow \boxed{R \ddot{\Delta \lambda} + 2 \sqrt{\frac{\mu}{R^3}} R \dot{\Delta r} = \frac{T_v}{m}} \quad (8) - 2.$$

$$iii) (R + \Delta r) \dot{\Delta \lambda} + 2 (\dot{r} + \dot{\Delta r}) (\Delta \lambda) + (R + \Delta r) \left(\sqrt{\frac{\mu}{R^3}} + \Delta v \right)^2 \frac{\sin \Delta \lambda \cos \Delta \lambda}{\Delta \lambda} = \frac{T_\lambda}{m}$$

$\sin \Delta \lambda \approx \Delta \lambda$, $\cos \Delta \lambda \approx 1$

$$\Rightarrow \boxed{R \dot{\Delta \lambda} + \left(\frac{\mu}{R^3} \right) R \Delta \lambda = \frac{T_\lambda}{m}} \quad (8) - 3 \quad \text{Decoupled!}$$

~~Clohesy-Wiltshire Equation. (= Circular Hill Dynamics)~~
~~= Hill-Clohesy-Wiltshire Eqn.)~~
 (8) - 1 ~ 2 = In-plane perturbation dyn.
 (8) - 3 = out-of-plane " "

IV) Using $\frac{\mu m}{R^2} = m r \omega_0^2 \Rightarrow \omega_0^2 = \frac{\mu}{R^3}$

In-plane dynamics

$$\begin{cases} \Delta \ddot{r} - 2\omega_0 R \Delta \dot{v} - 3\omega_0^2 \Delta r = \frac{F_r}{m} \\ R \Delta \ddot{v} + 2\omega_0 \Delta \dot{r} = \frac{F_v}{m} \end{cases}$$

Out-of-plane dynamics

$$R \Delta \ddot{\lambda} + \omega_0^2 R \Delta \lambda = \frac{F_\lambda}{m} \quad \Leftarrow \text{decoupled!}$$

V) Summary: Clohessy-Wiltshire Eqn = Circular Hill Dynamics
= Hill-Clohessy-Wiltshire Eqn

Define $x = \Delta r$, $y = R \Delta v$, $z = R \Delta \lambda$.

⇒ (Note (x, y, z) are perturbed quantities w.r.t. the ref. solution)

①

$$\begin{cases} \ddot{x} - 2\omega_0 \dot{y} - 3\omega_0^2 x = \frac{F_r}{m} \\ \ddot{y} + 2\omega_0 \dot{x} = \frac{F_v}{m} \\ \ddot{z} + \omega_0^2 z = \frac{F_\lambda}{m} \end{cases}$$

②

$$\begin{bmatrix} \ddot{x} \\ \ddot{y} \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & -2\omega_0 \\ 2\omega_0 & 0 \end{bmatrix}}_{\substack{\text{skew-symmetric} \\ \text{gyroscopic terms}}} \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} + \begin{bmatrix} -3\omega_0^2 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{F_r}{m} \\ \frac{F_v}{m} \end{bmatrix}$$

Diagram illustrating the state vector and forces:

$$\begin{bmatrix} \ddot{x} \\ \dot{x} \\ \ddot{y} \\ \dot{y} \\ \ddot{z} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} V_x \\ 2\omega_0 V_y + 3\omega_0^2 x + \frac{F_r}{m} \\ V_y \\ \ddots \end{bmatrix}$$

(Note: The diagram shows a state vector on the left and a corresponding force/acceleration vector on the right, with a large 'X' drawn over the entire diagram.)

③ Let $[x \ y \ z \ v_x \ v_y \ v_z] = [\text{or } R_{OV} \ R_{O\lambda} \ \dot{x} \ \dot{y} \ \dot{z}]$

In-plane Dynamics

$$\begin{bmatrix} \ddot{x} \\ \dot{v}_x \\ \ddot{y} \\ \dot{v}_y \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ 3\omega_0^2 & 0 & 0 & 2\omega_0 \\ 0 & 0 & 0 & 1 \\ 0 & -2\omega_0 & 0 & 0 \end{bmatrix}}_{\equiv A} \begin{bmatrix} x \\ v_x \\ y \\ v_y \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & 0 \\ \frac{1}{m} & 0 \\ 0 & 0 \\ 0 & \frac{1}{m} \end{bmatrix}}_{\equiv B} \begin{bmatrix} F_r \\ F_t \end{bmatrix}$$

Out-of-Plane

$$\begin{bmatrix} \ddot{z} \\ \dot{v}_z \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\omega_0^2 & 0 \end{bmatrix} \begin{bmatrix} z \\ v_z \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} F_n$$

(Introduce the concept of Stability, Controllability, & Observability)

② Free Orbital Motion ($\vec{F} \equiv 0, F_r = F_t = F_n \equiv 0$)

① Stability

In-plane Eigenvalue: $(0, 0) \pm j\omega_0$
 Out-of-plane " : $\pm j\omega_0$

perturbation diverges linearly ✓

③ (Homogeneous) Solution by State Transition Matrix (STM, Φ)

$$\Phi(t) = e^{At}$$

$$\frac{d}{dt} \Phi(t) = A\Phi(t), \Phi(0) = I.$$

$$\begin{bmatrix} x(t) \\ v_x(t) \\ y(t) \\ v_y(t) \end{bmatrix} = \begin{bmatrix} 4-3\cos\omega_0 t & \frac{\sin\omega_0 t}{\omega_0} & 0 & \frac{2(1-\cos\omega_0 t)}{\omega_0} \\ 3\omega_0 \sin\omega_0 t & \cos\omega_0 t & 0 & 2\sin\omega_0 t \\ 6\sin\omega_0 t - 6\omega_0 t & \frac{2(-1+\cos\omega_0 t)}{\omega_0} & 1 & \frac{4\sin\omega_0 t}{\omega_0} - 3t \\ 6\omega_0(-1+\cos\omega_0 t) & -2\sin\omega_0 t & 0 & -3+4\cos\omega_0 t \end{bmatrix} \begin{bmatrix} x(0) \\ v_x(0) \\ y(0) \\ v_y(0) \end{bmatrix}$$

② ! Contains Secular Terms

in contrast to Out-of-plane Dyn

② General Solution to Linear System

$$\begin{aligned} \textcircled{1} \quad & \begin{cases} \dot{x} = Ax + Bu \\ y = Cx \end{cases} \\ & \begin{cases} x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)} Bu(\tau) d\tau \\ y(t) = C e^{At}x(0) + C \int_0^t e^{A(t-\tau)} Bu(\tau) d\tau \end{cases} \end{aligned}$$

② Transfer Func.

$$G(s) = \frac{Y(s)}{U(s)} = C(sI - A)^{-1}B$$

ex) TF for out-of-phase dynamics

$$\begin{aligned} \ddot{z} + \omega_s^2 z &= \frac{F_A}{m} \\ \Rightarrow (s^2 + \omega_s^2) z &= \frac{F_A}{m} \quad (\text{IC} \equiv 0 \text{ by def.}) \\ \Rightarrow z &= \frac{1/m}{s^2 + \omega_s^2} F_A \end{aligned}$$

③ Completely Controllable iff $\text{rank}[B \ AB \ \dots \ A^{n-1}B] = n$

ex) Controllability of out-of-phase Dynamics

$$A = \begin{bmatrix} 0 & 1 \\ -\omega_s^2 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\text{rank}[B \ AB] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = 2 \Rightarrow \text{controllable}$$

④ Completely observable iff $\text{rank} \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix} = n$

cf. Duality between controllability & observability

② Orbital Maneuver Problems (as control problems)

1. Rendezvous: State-to-State Transfer in a fixed time
BC at terminal time t_f

1. Rendezvous

i) State-to-State Transfer (~~in a fixed time~~)

ii) BC at terminal time t_f $\begin{cases} 4 \text{ in-plane} \\ 2 \text{ out-of-plane} \end{cases}$

2. Orbit Transfer

i) State-to-(Reduced) State transfer (~~in a fixed time~~)

ii) BC at ~~be~~ t_f $\begin{cases} 3 \text{ in-plane} \\ 2 \text{ out-of-plane} \end{cases}$

iii) 'don't care' about $y(t_f)$: angular disp.

3. Interception Problem

i) State-to-Position vector transfer

ii) BC at time t_f $\begin{cases} 2 \text{ in-plane} \\ 1 \text{ out-of-plane} \end{cases}$

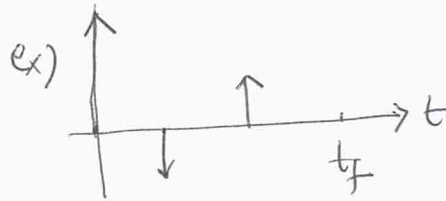
iii) Don't care about velocity.

② Modeling of Impulsive (Propulsion) Forces

⊕ ~~High~~ ⊕

1. high thrust assumption → impulsive model

$$\begin{bmatrix} F_r \\ F_v \\ F_\lambda \end{bmatrix} = \sum_{i=1}^{N_z} \begin{bmatrix} I_i^r \\ I_i^v \\ I_i^\lambda \end{bmatrix} \delta(t-t_i), \quad 0 \leq t_1 \leq \dots \leq t.$$



2. Impulse [Instantaneous change of velocity
No change of position $\Leftrightarrow \int_t^{t+\epsilon} v dt \approx 0$.

3. $\Delta \vec{V}_i = \vec{V}(t_i^+) - \vec{V}(t_i^-)$

③ Rendezvous Problem

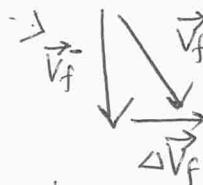
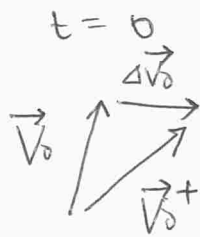
~~1. Two - Impulse Rendezvous Problem. Maneuver (Finite Dimensional Problem)~~

~~⊗ ~~Given~~: $(x_0, \dot{x}_0, y_0, \dot{y}_0, z_0, \dot{z}_0)$~~

$\Rightarrow (x_f, \dot{x}_f, y_f, \dot{y}_f, z_f, \dot{z}_f)$ on $[0, t_f]$

1. Two - Impulse Maneuver (Finite Dimensional Problem)

1)



1) $[t_0^+, t_f^-]$ is a 'coast phase'

III) In-plane Maneuver.

a) $t_0 \rightarrow t_0^+$

$$K \begin{bmatrix} x_0 \\ \dot{x}_0 \\ y_0 \\ \dot{y}_0 \end{bmatrix} \xrightarrow{\Delta \vec{V}_0} \begin{bmatrix} x_0 \\ \dot{x}_0^+ \\ y_0 \\ \dot{y}_0^+ \end{bmatrix}$$

b) Coasting. $[t_0^+, t_f^-]$

$$\begin{array}{l} \text{Known} \\ \text{Unknown} \\ \text{Known} \\ \text{Unknown} \end{array} \begin{bmatrix} x_f \\ \dot{x}_f^- \\ y_f \\ \dot{y}_f^- \end{bmatrix} = \underbrace{e^{A t_f}}_{\text{STM}} \begin{bmatrix} x_0 \\ \dot{x}_0^+ \\ y_0 \\ \dot{y}_0^+ \end{bmatrix} = \Phi \begin{bmatrix} x_0 \\ \dot{x}_0^+ \\ y_0 \\ \dot{y}_0^+ \end{bmatrix} \begin{array}{l} \text{Known} \\ \text{Unknown} \\ \text{Known} \\ \text{Unknown} \end{array}$$

c) $t_f^- \rightarrow t_f$

$$\begin{bmatrix} t_f^- \\ x_f \\ \dot{x}_f^- \\ y_f \\ \dot{y}_f^- \end{bmatrix} \xrightarrow{\Delta \vec{V}_f} \begin{bmatrix} t_f \\ x_f \\ \dot{x}_f \\ y_f \\ \dot{y}_f \end{bmatrix}$$

IV) Out-of-Plane Maneuver

a) $t_0 \rightarrow t_0^+$

$$\begin{bmatrix} z_0 \\ \dot{z}_0 \end{bmatrix} \xrightarrow{\Delta \vec{V}_0} \begin{bmatrix} z_0 \\ \dot{z}_0^+ \end{bmatrix}$$

b) Coasting $[t_0^+, t_f^-]$

$$\begin{array}{l} \text{Unknown} \end{array} \begin{bmatrix} z_f \\ \dot{z}_f^- \end{bmatrix} = \Phi \begin{bmatrix} z_f \\ \dot{z}_f^+ \end{bmatrix} \begin{array}{l} \text{Unknown} \end{array}$$

c) $t_f^- \rightarrow t_f$

$$\begin{bmatrix} z_f \\ \dot{z}_f^- \end{bmatrix} \rightarrow \begin{bmatrix} z_f \\ \dot{z}_f \end{bmatrix}$$

V) Total Maneuver

$$\Delta \vec{V}_0 = (\dot{x}_0^+ - \dot{x}_0) \vec{e}_r + (\dot{y}_0^+ - \dot{y}_0) \vec{e}_v + (\dot{z}_0^+ - \dot{z}_0) \vec{e}_\lambda$$

$$\Delta \vec{V}_f = (\dot{x}_f - \dot{x}_f^-) \vec{e}_r + (\dot{y}_f - \dot{y}_f^-) \vec{e}_v + (\dot{z}_f - \dot{z}_f^-) \vec{e}_\lambda$$

2. Piecewise Constant Maneuver. (Finite Dimensional Problem)

I) $\vec{F}(t) = [F_r(t) \ F_v(t) \ F_\lambda(t)]^T$

$$= \begin{cases} [F_r^1 \ F_v^1 \ F_\lambda^1]^T, & 0 \leq t \leq t_1 \\ [F_r^2 \ F_v^2 \ F_\lambda^2]^T, & t_1 \leq t \leq t_f \end{cases}$$

ii) ~~In-plane Maneuver~~

~~$$\begin{bmatrix} \Delta x_f \\ \Delta \dot{x}_f \\ \Delta y_f \\ \Delta \dot{y}_f \end{bmatrix} = e^{A t_f} \begin{bmatrix} \dots \end{bmatrix}$$~~

ii) In-plane maneuver

$$\begin{bmatrix} x_f \\ \dot{x}_f \\ y_f \\ \dot{y}_f \end{bmatrix} = e^{A t_f} \begin{bmatrix} x_0 \\ \dot{x}_0 \\ y_0 \\ \dot{y}_0 \end{bmatrix} + \int_0^{t_f} e^{A(t_f-z)} B \begin{bmatrix} F_r(z) \\ F_v(z) \\ \cancel{F_\lambda(z)} \end{bmatrix} dz$$

$$= \dots + \int_0^{t_1} e^{A(t_f-z)} B dz \begin{bmatrix} F_r^1 \\ F_v^1 \end{bmatrix} + \int_{t_1}^{t_f} e^{A(t_f-z)} B dz \begin{bmatrix} F_r^2 \\ F_v^2 \end{bmatrix}$$

iii) Out-of-Plane Maneuver

$$\begin{bmatrix} z_f \\ \dot{z}_f \end{bmatrix} = e^{A t_f} \begin{bmatrix} z_0 \\ \dot{z}_0 \end{bmatrix} + \int_0^{t_1} e^{A(t_f-t)} B dz F_\lambda + \int_{t_1}^{t_f} e^{A(t_f-t)} B dz F_\lambda$$

3. Optimal Rendezvous Maneuver (Infinite Dimensional)

i) In-Plane Maneuver

$$\begin{bmatrix} \bar{r}(t) \\ \bar{v}(t) \end{bmatrix} = B^T e^{A(t-t_f)} W(0, t_f)^{-1} \left\{ \begin{bmatrix} x_f \\ \dot{x}_f \\ y_f \\ \dot{y}_f \end{bmatrix} - e^{At_f} \begin{bmatrix} x_0 \\ \dot{x}_0 \\ y_0 \\ \dot{y}_0 \end{bmatrix} \right\}$$

$$\text{where } W(0, t_f) = \int_0^{t_f} e^{A(t_f-\tau)} B B^T e^{A^T(t_f-\tau)} d\tau$$

ii) Out-of-Plane Maneuver.

$$\bar{r}_1(t) = B^T e^{A(t-t_f)} W(0, t_f)^{-1} \left\{ \begin{bmatrix} z_f \\ \dot{z}_f \end{bmatrix} - e^{At_f} \begin{bmatrix} z_0 \\ \dot{z}_0 \end{bmatrix} \right\}$$

iii) This is optimal in the sense that

$$\min J = \int_0^{t_f} \left\| \begin{bmatrix} \bar{r}(z) \\ \bar{v}(z) \end{bmatrix} \right\|^2 dz \quad \& \quad J = \int_0^{t_f} \bar{r}_1(z)^2 dz$$

~~Verification by~~
Verification -

iv) Verification

① Compatibility with BC - direct substitution of u into system dynamics

② Optimality - after learning optimal control

② Orbit Transfer Problem.

Same as rendezvous except that no BC on $v(t_f)$ exists

~~Two-Impulse Maneuver.~~

$$(x_0, \dot{x}_0, y_0, \dot{y}_0, z_0, \dot{z}_0) \Rightarrow (x_0, \dot{x}_0, \boxed{0}, \boxed{0}, z_0, \dot{z}_0)$$



1. 2-Impulse Transfer. (Finite Dimensional)

1) In-Plane Maneuver

$$\textcircled{a} \begin{matrix} t_0 & & t_0^+ \\ \begin{bmatrix} x_0 \\ \dot{x}_0 \\ y_0 \\ \dot{y}_0 \end{bmatrix} & \xrightarrow{\Delta \vec{V}_0} & \begin{bmatrix} x_0 \\ \dot{x}_0^+ \\ y_0 \\ \dot{y}_0^+ \end{bmatrix} \end{matrix}$$

~~Coast~~ $[t_0^+, t_f^-]$

$$\begin{bmatrix} x_f^- \\ \dot{x}_f^- \\ y_f^- \\ \dot{y}_f^- \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\equiv C} e^{A t_f} \begin{bmatrix} x_0 \\ \dot{x}_0^+ \\ y_0 \\ \dot{y}_0^+ \end{bmatrix}$$

3 eqns for 4 unknowns: 1 redundancy.

$$\textcircled{c} \begin{matrix} t_f^- & & t_f \\ \begin{bmatrix} x_f^- \\ \dot{x}_f^- \\ y_f^- \\ \dot{y}_f^- \end{bmatrix} & \rightarrow & \begin{bmatrix} x_f \\ \dot{x}_f \\ y_f \\ \dot{y}_f \end{bmatrix} \end{matrix}$$

ii) Out-of-Plane Maneuver.

= Same as Rendezvous Maneuver.

iii) Total Maneuver

$$\begin{cases} \Delta \vec{V}_0 = (\dot{x}_0^+ - \dot{x}_0) \vec{e}_r + (\dot{y}_0^+ - \dot{y}_0) \vec{e}_v + (\dot{z}_0^+ - \dot{z}_0) \vec{e}_\lambda \\ \Delta \vec{V}_f = (\dot{x}_f - \dot{x}_f^-) \vec{e}_r + (\dot{y}_f - \dot{y}_f^-) \vec{e}_v + (\dot{z}_f - \dot{z}_f^-) \vec{e}_\lambda \end{cases}$$

2. Optimal Orbit Transfer Maneuver (Infinite Dimensional)

i) In-Plane Maneuver

$$\begin{bmatrix} F_r(t) \\ F_t(t) \end{bmatrix} = B^T e^{A^T(t_f - \tau)} C^T W(0, t_f)^{-1} \begin{bmatrix} x_f \\ \dot{x}_f \\ y_f \end{bmatrix} - (e^{A t_f} \begin{bmatrix} x_0 \\ \dot{x}_0 \\ y_0 \\ \dot{y}_0 \end{bmatrix})$$

$$\text{Where } W(0, t_f) = C \int_0^{t_f} e^{A(t_f - \tau)} B B^T e^{A^T(t_f - \tau)} d\tau C^T$$

ii) Out-of-Plane Maneuver.

= Same as ~~the~~ Rendezvous Maneuver.

iii) Optimal in the sense that.

$$\min J = \int_0^{t_f} \left\| \begin{bmatrix} F_r(t) \\ F_t(t) \end{bmatrix} \right\|^2 dt, \quad J = \int_0^{t_f} T_A(t) dt$$

3. 2-Impulse Orbit Transfer between Coplanar Circular Orbits

~~Initial Radius~~

~~Initial / Final Radius ; R_1 / R_2 ($R_1 < R_2$)~~

~~Linearize about the circular orbit with radius R_2~~

i) Boundary Conditions

$$\begin{bmatrix} x_0 = R_1 - R_2 < 0 \\ \dot{x}_0 = 0 \\ y_0 = 0 \text{ w.l.o.g.} \\ \dot{y}_0 = -\frac{3}{2} \omega_0 x_0 \end{bmatrix}$$

$$\dot{x}_0 = 0$$

$$y_0 = 0 \text{ w.l.o.g.}$$

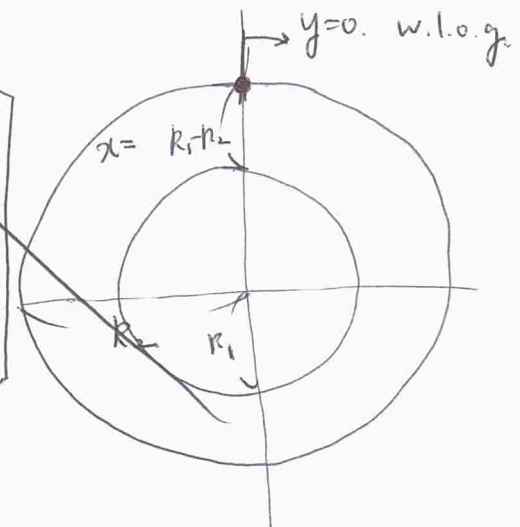
$$\dot{y}_0 = -\frac{3}{2} \omega_0 x_0$$

$$\begin{bmatrix} x_f = 0 \\ \dot{x}_f = 0 \\ y_f = \text{free} \\ \dot{y}_f = 0 \end{bmatrix}$$

$$\dot{x}_f = 0$$

$$y_f = \text{free}$$

$$\dot{y}_f = 0$$



~~From the CW-eqn.~~

$$\ddot{x}_0 - 2\omega_0 \dot{y}_0 - 3\omega_0^2 x_0 = \frac{F_r}{m} \text{ at } t=0$$

$$\Rightarrow \dot{y}_0 = -\frac{3}{2} \omega_0 x_0 \quad (\text{roughly } \dot{y}_0 \approx r_1 v_1 - r_2 v_2)$$

3. 2-Impulse Orbit Transfer between Coplanar Circular Orbits

Initial/Final Radius: R_1/R_2 ($R_1 < R_2$)

Linearize about the circular ~~at~~ orbit with radius R_2

i) Boundary Conditions

$$\begin{array}{l} t=t_0 \\ \left[\begin{array}{l} x_0 = R_1 - R_2 < 0 \\ \dot{x}_0 = 0 \\ y_0 = 0 \text{ w.l.o.g.} \\ \dot{y}_0 = -\frac{3}{2}\omega_0 x_0 \end{array} \right] \end{array} \quad \begin{array}{l} t=t_f \\ \left[\begin{array}{l} x_f = 0 \\ \dot{x}_f = 0 \\ y_f = \text{free} \\ \dot{y}_f = 0 \end{array} \right] \end{array}$$

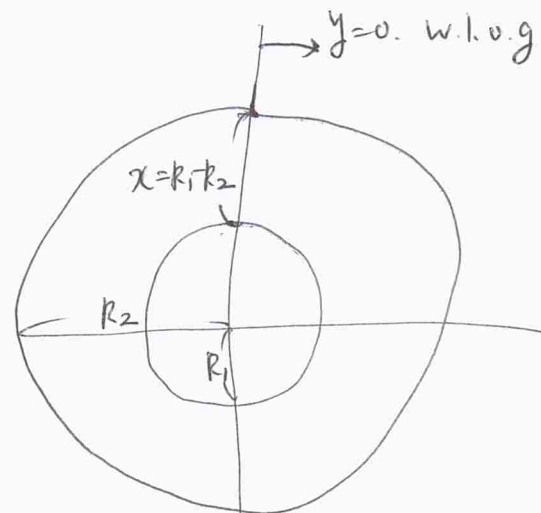
$\Downarrow$

From CW-egns. at $t=0$

$$\ddot{x}_0 - 2\omega_0 \dot{y}_0 - 3\omega_0^2 x_0 = \frac{F_r}{m}$$

$$\Rightarrow \dot{y}_0 = -\frac{3}{2}\omega_0 x_0$$

$$(\text{Roughly } \dot{y}_0 \approx r_1 V_1 - r_2 V_2)$$



ii) Assume $\begin{cases} \Delta \vec{V}_0 & \text{is tangential} \\ \Delta \vec{V}_f & \text{is both radial \& tangential} \end{cases}$

iii)

$$\begin{array}{c} t_0 \quad \rightarrow \quad t_0^+ \\ \left[\begin{array}{c} R_1 - R_2 \\ 0 \\ 0 \\ -\frac{3}{2}\omega_0 x_0 \end{array} \right] \rightarrow \left[\begin{array}{c} R_1 - R_2 \\ 0 \\ 0 \\ -\frac{3}{2}\omega_0 x_0 + \Delta V_{y0} \end{array} \right] \end{array} \quad \begin{array}{l} \leftarrow \text{no radial burn} \\ \text{to be determined} \end{array}$$

(19)

IV) Using STM,

$$\textcircled{1} x(t_f) = x_f = 0 = (4 - 3\cos\omega_0 t_f) x_0 + \frac{2(1 - \cos\omega_0 t_f)}{\omega_0} \left[-\frac{3}{2}\omega_0 x_0 + \Delta V_{y0} \right]$$

$$\Rightarrow \Delta V_{y0} = -\frac{\omega_0 x_0}{2(1 - \cos\omega_0 t_f)}$$

$$\Rightarrow \Delta \vec{V}_0 = -\frac{\omega_0 x_0}{2(1 - \cos\omega_0 t_f)} \vec{e}_v$$

$$\begin{aligned} \textcircled{2} \dot{x}(t_f) &= 3\omega_0 \sin\omega_0 t_f x_0 + 2\sin\omega_0 t_f \left[-\frac{3}{2}\omega_0 x_0 - \frac{\omega_0 x_0}{2(1 - \cos\omega_0 t_f)} \right] \\ &= -\frac{\sin\omega_0 t_f}{1 - \cos\omega_0 t_f} \omega_0 x_0 \end{aligned}$$

$$\begin{aligned} \textcircled{3} \dot{y}(t_f) &= 6\omega_0(-1 + \cos\omega_0 t_f)x_0 + (-3 + 4\cos\omega_0 t_f) \\ &\quad \times \left[-\frac{3}{2}\omega_0 x_0 - \frac{\omega_0 x_0}{2(1 - \cos\omega_0 t_f)} \right] \\ &= -\frac{\cos\omega_0 t_f}{2(1 - \cos\omega_0 t_f)} \omega_0 x_0 \end{aligned}$$

$$\begin{aligned} \Delta \vec{V}_f &= \cancel{\dot{x}_f} (\dot{x}_f - \dot{x}_f^-) \vec{e}_r + (\dot{y}_f - \dot{y}_f^-) \vec{e}_v \\ &= \frac{\sin\omega_0 t_f}{1 - \cos\omega_0 t_f} \omega_0 x_0 \vec{e}_r + \frac{\cos\omega_0 t_f}{2(1 - \cos\omega_0 t_f)} \omega_0 x_0 \vec{e}_v \end{aligned}$$

V) ~~Special Case~~
 $x_0 = 0 \Rightarrow \Delta \vec{V}_0$

VI) Special Case I

$$x_0 = 0 \Rightarrow \Delta \vec{V}_0 = \Delta \vec{V}_f = 0 \quad (\text{trivial})$$

Special Case II

$$t_f = \frac{\pi}{\omega_0} = (\text{half period})$$

$$\begin{cases} \Delta \vec{V}_0 = -\frac{\omega_0 \alpha_0}{4} \vec{e}_v \\ \Delta \vec{V}_f = 0 \cdot \vec{e}_r - \frac{\omega_0 \alpha_0}{4} \vec{e}_v \end{cases} \quad (\text{tangential})$$

$\Rightarrow$ Hohmann Transfer.

Matrix Exponential & Integration

$$i) e^{At} = I + At + \frac{1}{2!}(At)^2 + \dots$$

$$\dot{\Phi}(t) = A\Phi(t), \quad \Phi(0) = I.$$

$$ii) \psi_1(t) \triangleq \int_0^t e^{A(t-\tau)} B d\tau$$

$$\frac{d\psi_1}{dt} = e^{A(t-t)} B, \quad \psi_1(0) = 0$$

$$\frac{d\psi_1}{dt} = \Phi(t-t)B$$

$$iii) \psi_2(t) \triangleq \int_{t_1}^t e^{A(t-\tau)} B d\tau$$

$$\frac{d\psi_2}{dt}(t) = B + A \int_{t_1}^t e^{A(t-\tau)} B d\tau$$

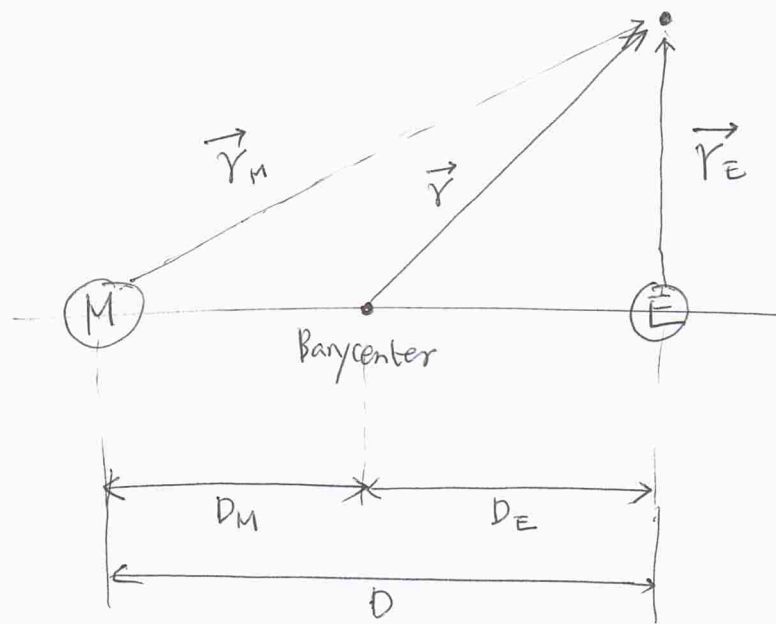
$$\frac{d\psi_2}{dt}(t) = B + A\psi_2(t), \quad \psi_2(t_1) = 0$$

$$iv) W(t) \triangleq \int_0^t e^{A(t-\tau)} B B^T e^{A^T(t-\tau)} d\tau$$

$$\begin{aligned} \frac{dW(t)}{dt} &= B B^T + \int_0^t A e^{A(t-\tau)} B B^T e^{A^T(t-\tau)} d\tau \\ &\quad + \int_0^t e^{A(t-\tau)} B B^T e^{A^T(t-\tau)} A^T d\tau \end{aligned}$$

$$\Rightarrow \frac{dW(t)}{dt} = B B^T + A W(t) + W(t) A^T, \quad W(0) = 0$$

< Restricted 3-Body Model >



© Basic Properties

1. Distance D between Earth & Moon. ~~ω_0~~

$$\omega_0 = \sqrt{\frac{G(M_M + M_E)}{D^3}} \Rightarrow D = \sqrt[3]{\frac{G(M_M + M_E)}{\omega_0^2}}$$

2. Rotation rate of barycentric coordinate frame is $\vec{\omega} = \omega_0 \vec{k}$

3. Earth-Moon data

i) $\mu_E = 398,601 \text{ km}^3/\text{s}^2$

ii) $\mu_M = 4.887 \text{ km}^3/\text{s}^2$

iii) $D = 384,748 \text{ km}$

$D_E = 4,674 \text{ km}$

$D_M = 380,073 \text{ km}$

iv) $\omega_0 = 2.66199 \times 10^{-6} \text{ rad/sec.}$

② EOM

$$m\ddot{\vec{r}} = \vec{F}_E + \vec{F} \quad \textcircled{1}$$

$$\Rightarrow \ddot{\vec{r}} = -\frac{\mu_E}{r_E^3} \vec{r}_E - \frac{\mu_M}{r_M^3} \vec{r}_M + \frac{\vec{F}}{m} \quad \textcircled{1}$$

$$\begin{cases} -\frac{\mu_E}{r_E^3} \vec{r}_E = -\frac{\mu_E(x-D_E)}{r_E^3} \vec{i} - \frac{\mu_E y}{r_E^3} \vec{j} - \frac{\mu_E z}{r_E^3} \vec{k} \\ -\frac{\mu_M}{r_M^3} \vec{r}_M = -\frac{\mu_M(x+D_M)}{r_M^3} \vec{i} - \frac{\mu_M y}{r_M^3} \vec{j} - \frac{\mu_M z}{r_M^3} \vec{k} \\ \vec{F} = F_x \vec{i} + F_y \vec{j} + F_z \vec{k} \end{cases}$$

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$\begin{aligned} \Rightarrow \dot{\vec{r}} &= \dot{x}\vec{i} + \dot{y}\vec{j} + \dot{z}\vec{k} + \vec{\omega} \times (x\vec{i} + y\vec{j} + z\vec{k}) \\ &= (\dot{x} - y\omega_0)\vec{i} + (\dot{y} + x\omega_0)\vec{j} + \dot{z}\vec{k} \end{aligned}$$

$$\Rightarrow \ddot{\vec{r}} = (\ddot{x} - 2\omega_0 \dot{y} - \omega_0^2 x)\vec{i} + (\ddot{y} + 2\omega_0 \dot{x} - \omega_0^2 y)\vec{j} + \ddot{z}\vec{k} \quad \textcircled{2}$$

$$\textcircled{1} = \textcircled{2}$$

$$\begin{cases} \ddot{x} - 2\omega_0 \dot{y} - \omega_0^2 x = -\frac{\mu_E(x-D_E)}{r_E^3} - \frac{\mu_M(x+D_M)}{r_M^3} + \frac{F_x}{m} \\ \ddot{y} + 2\omega_0 \dot{x} - \omega_0^2 y = -\frac{\mu_E y}{r_E^3} - \frac{\mu_M y}{r_M^3} + \frac{F_y}{m} \\ \ddot{z} = -\frac{\mu_E z}{r_E^3} - \frac{\mu_M z}{r_M^3} + \frac{F_z}{m} \end{cases}$$

$$r_E^2 = (x-D_E)^2 + y^2 + z^2$$

$$r_M^2 = (x+D_M)^2 + y^2 + z^2$$

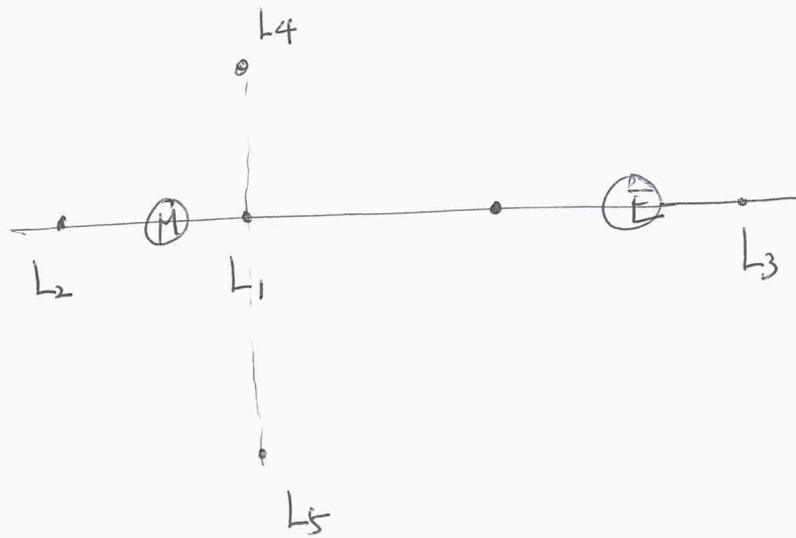
① (Unforced) Equilibrium Solutions = Libration Points

i) In case $\vec{F} \equiv 0$, $\exists$ 5 equilibrium solutions ($L_1, \dots, L_5$)

ii) Earth-Moon Libration Points

	$\bar{x}$	$\bar{y}$	$\bar{z}$
L_1	-321,880.3	0	0
L_2	-444,645.6	0	0
L_3	386,694.8	0	0
L_4	-187,699.3	333,201.5	0
L_5	-187,699.3	-333,201.5	0

} Colinear
Lagrangian
Points



② Linearization about libration points

$$\begin{cases} x = \bar{x} + \Delta x \\ y = \bar{y} + \Delta y \\ z = \bar{z} + \Delta z \end{cases}$$

1. Linearization about $L_1 \sim L_3$

$$\begin{aligned} 1) \quad \Delta \ddot{x} - 2\omega_0 \Delta \dot{y} + C_1 \Delta x &= \frac{F_x}{m} \\ \Delta \ddot{y} + 2\omega_0 \Delta \dot{x} + C_2 \Delta y &= \frac{F_y}{m} \\ \Delta \ddot{z} + C_3 \Delta z &= \frac{F_z}{m} \end{aligned}$$

	C_1	C_2	C_3
L_1	$-11.2954\omega_0^2$	$4.1477\omega_0^2$	$5.1477\omega_0^2$
L_2	$-7.3809\omega_0^2$	$2.1904\omega_0^2$	$3.1904\omega_0^2$
L_3	$-3.0214\omega_0^2$	$0.0107\omega_0^2$	$1.0107\omega_0^2$

ii) Eigenvalues

	In-Plane	Out-of-Plane
L_1	$\pm 2.9321\omega_0, \pm j 2.3344\omega_0$	$\pm j 2.2689\omega_0$
L_2	$\pm 2.1587\omega_0, \pm j 1.8627\omega_0$	$\pm j 1.7882\omega_0$
L_3	$\pm 0.1779\omega_0, \pm j 1.0104\omega_0$	$\pm j 1.0053\omega_0$

2. Linearization about $L_4 \sim L_5$

$$\begin{aligned} 1) \quad \Delta \ddot{x} - 2\omega_0 \Delta \dot{y} - 0.75\omega_0^2 \Delta x + 1.2674\omega_0^2 \Delta y &= \frac{F_x}{m} \\ \Delta \ddot{y} + 2\omega_0 \Delta \dot{x} + \cancel{1.2674} \omega_0^2 \Delta x - 2.25\omega_0^2 \Delta y &= \frac{F_y}{m} \\ &+ 1.27 \\ \Delta \ddot{z} + \omega_0^2 \Delta z &= \frac{F_z}{m} \end{aligned}$$

ii) In-plane Eigenvalues : $\pm j 0.9545\omega_0, \pm j 0.2982\omega_0$
 Out-of-Plane " : $\pm j \omega_0$

iii) Cannot conclude that the original nonlinear dynamics is stable (by the Lyapunov indirect method)

In fact, the original nonlinear system is aperiodic
= almost periodic

© ~~Halo Orbits~~ Periodic Orbits

1. Halo Orbit (Original Definition in Celestial Mechanics)

- Larger, Periodic orbit that is a solution of nonlinear diff. eqn. of motion

2. Quasi-Periodic Orbit.

i) Lissajous trajectory: In-plane and Out-of-Plane frequencies are different and their ratio is ~~not~~ irrational (The trajectory does NOT close.)

ii) Halo orbit: In-plane and Out-of-Plane frequencies are rational (The trajectory does close and is periodic in itself.)

[However if a spacecraft is controlled, it does not close due to tracking error. That's why the halo orbit is called quasi-periodic though it is periodic in theory]

of. Often ~~Lissajous Halo orbit~~ the def. of Halo orbit includes the Lissajous trajectory

② Halo Orbit

1. Halo Orbit about L_3

(In-plane: Periodic with $\omega = 1.0104\omega_0$
 Out-of-plane: " $\omega = 1.0053\omega_0$

$\Rightarrow$ Almost Periodic, but NOT rationally related

2. What are the initial conditions that compose a Halo Orbit at a single frequency?

1) Let $\lambda = \pm j\omega$, $(\xi_0, \bar{\xi}_0)$ are eigenvalues and eigenvectors of the system matrix A .

Then, the solution to ~~$\Delta\dot{x} = A\Delta x$, $\Delta x(0) = \Delta x_0$~~

~~can be~~ $\Delta\dot{x} = A\Delta x$, $\Delta x(0) = \Delta x_0$

can be expanded in eigenspace as

$$\Delta x(t) = C e^{At} \Delta x_0$$

$$= a_1 \xi_0 e^{j\omega t} + a_2 \bar{\xi}_0 e^{-j\omega t}, \quad (a_1, a_2) \in \mathbb{R}$$

$$\Rightarrow \Delta x(t) = a_1 \xi_0 (\cos \omega t + j \sin \omega t)$$

$$+ a_2 \bar{\xi}_0 (\cos \omega t - j \sin \omega t)$$

$$= (a_1 \xi_0 + a_2 \bar{\xi}_0) \cos \omega t + j(a_1 \xi_0 - a_2 \bar{\xi}_0) \sin \omega t$$

$$\Rightarrow \begin{cases} \Delta x(0) = a_1 \xi_0 + a_2 \bar{\xi}_0 \\ \Delta \dot{x}(0) = j\omega(a_1 \xi_0 - a_2 \bar{\xi}_0) \end{cases}$$

$\Rightarrow$ Solve for (a_1, a_2)

1) If $x(0) \in \text{span}\{\text{Re } \xi, \text{Im } \xi\}$, then the solution is periodic with frequency ω .

$$\begin{aligned}
 1) \quad & \left\{ \begin{array}{l} \text{If } a_1 = a_2, \Rightarrow \Delta x(0) = 2a_1 \operatorname{Re} \xi \\ \text{If } a_1 = -a_2 \Rightarrow \Delta x(0) = 2a_1 \operatorname{Im} \xi \end{array} \right. \\
 & \Rightarrow x(t) \text{ is periodic with frequency } \omega
 \end{aligned}$$

$$1) \quad \cancel{\text{If } \Delta x(0) \in \{ \operatorname{Re} \xi, \operatorname{Im} \xi \}}$$

$$\begin{aligned}
 1) \quad & \text{If } \Delta x(0) \in \operatorname{span} \{ \operatorname{Re} \xi, \operatorname{Im} \xi \}, \text{ then the solution} \\
 & \text{is periodic with frequency } \omega
 \end{aligned}$$